ST3189 MACHINE LEARNING COURSEWORK

REPORT

SRN:

Executive Summary

This research project analyses two datasets: the **Body Fat Prediction Dataset** and the **Heart Failure Clinical Records Dataset**, which will be used to identify and answer research questions related to each respective dataset.

In this day and age where obesity and cardiovascular diseases pose significant health risks, identifying patterns leading to these conditions are crucial. Accurately predicting body fat percentage and identifying factors influencing mortality of heart failure patients helps in early intervention, personalised treatment plans, and preventive healthcare strategies. This enables individuals to make informed lifestyle choices and adopt healthier habits. On the other hand, healthcare providers can also optimise patient management and create individualised medical care. Moreover, the insights generated can contribute to policy development and effective resource allocation in order to mitigate the impact of these health conditions.

The **Body Fat Prediction Dataset**, which is supplied by Dr. A. Garth Fisher, provides an extensive overview of various body circumference measurements and body fat estimates of 252 men. With 15 variables and 252 observations, this dataset is a suitable resource to predict individuals' body fat percentage using their physical measurements.

This analysis uses unsupervised learning such as *Principal Component Analysis*, *K-Means Clustering* and *Hierarchical Clustering* to reduce dimensionality and identify potential clusters. Through those, common characteristics of clusters were discovered which can be valuable for further analysis. Additionally, regression techniques such as *Linear*, *Lasso*, *Ridge Regression*, *Decision Tree* and *Random Forest* were employed to predict the body fat percentage and assess model accuracy. Identification of significant predictors helped to improve model accuracy and supports early intervention, while leveraging the best model enhances decision making and minimises error.

The **Heart Failure Clinical Records Dataset**, which is sourced from medical records of heart failure patients, also provides an overview of clinical and lifestyle information. With 13 variables and 299 observations, this dataset is also a suitable resource to predict mortality based on the clinical and lifestyle information given.

This analysis uses classification techniques such as *Logistic Regression*, *K-nearest Neighbours*, *Decision Tree* and *Random Forest* to predict mortality and assess model accuracy. Through those, significant predictors were also identified in order to improve model accuracy. While all the models exhibited high accuracy scores, *Logistic Regression* had the best outcome amongst the rest and had the stronger ability to differentiate between survivors and non-survivors.

In conclusion, this research highlights the importance of clustering data, accurate model predictions and significant predictors in managing health conditions. By integrating these insights through data analysing, healthcare intervention and policies can be enhanced to strengthen an individual's overall welfare. Advancing in future research, it is essential to investigate different methods to improve predictive models, so that health conditions and preventive strategies can be even more well-managed and optimised.

TABLE OF CONTENTS

1. INTRODUCTION

Machine Learning involves analysing datasets via algorithms and statistical models which enables computers to analyse data, identify patterns and make predictions (What Is Machine Learning? Definition, Types, Tools & More, 2024). Machine Learning techniques will be applied on two datasets – Body Fat Prediction Dataset (Body Fat Prediction Dataset, 2021) and Heart Failure Clinical Records Dataset (Heart Failure Clinical Records, 2020). They will be referred to as body fat data and heart failure data respectively.

1.1 BACKGROUND

1.1.1 BODY FAT PREDICTION DATASET

This dataset was sourced from Kaggle, and it contains body composition measurements of 252 individuals with no duplicated or missing values. Its goal is to identify meaningful clusters using unsupervised learning, evaluate five regression models and to determine the most significant predictors of body fat percentage. The dataset includes 15 variables – Density, Age, Weight, Height, various circumferences (e.g. Hip, Forearm, Neck) and Body Fat as the target variable. [Appendix 8.1]

1.1.2 HEART FAILURE CLINICAL RECORDS DATASET

This dataset was sourced from UCI Repository, and it contains medical data of 299 patients who had heart failure with no duplicated or missing values. Its goal is to evaluate four classification models, and to determine which clinical and lifestyle features are essential in predicting mortality. The dataset includes 13 clinical and lifestyle variables – Age, Anaemia, Creatinine Phosphokinase, Diabetes, Ejection Fraction, High Blood Pressure, Platelets, Serum Creatinine, Serum Sodium, Sex, Smoking, Time and Death Event as the target variable. [Appendix 8.2]

1.2 SUBSTANTIVE ISSUES

1.2.1 BODY FAT PREDICTION DATASET

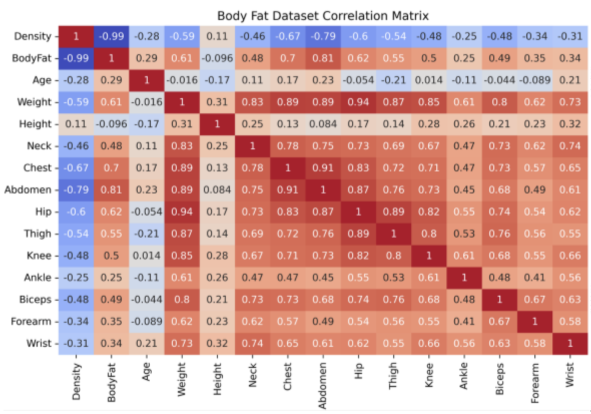
Body fat percentage is the amount of fat in the body divided by the total body weight, and it is one of the many ways that a person can determine if he is at risk of health problems (British Heart Foundation, n.d.). An excessive percentage of body fat can lead to several health issues such as type 2 diabetes and cardiovascular disease (Otis, 2025). By utilising the various body circumference measurements to predict body fat percentage, patterns can be identified which is useful in preventive healthcare. Early detection of high body fat percentage can lead individuals to adopt lifestyle changes.

1.2.2 HEART FAILURE CLINICAL RECORDS DATASET

Heart failure occurs when the heart muscle fails to pump efficiently, leading to a lack of delivery in oxygen and nutrient blood to the rest of the body (*What Is Heart Failure?*, 2023). It is a serious condition as it is unpredictable and limits the activities one can do. High mortality rate is also often associated with the condition (Website, 2024). Similarly, prediction of mortality with the clinical and lifestyle information can help to identify patients who are at a higher risk, thus leading to early intervention.

2. Exploratory Data Analysis

2.1 Body Fat Prediction Dataset



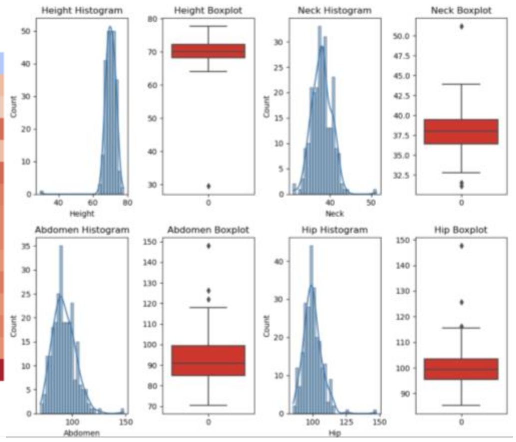


FIGURE 2.1.1: CORRELATION MATRIX

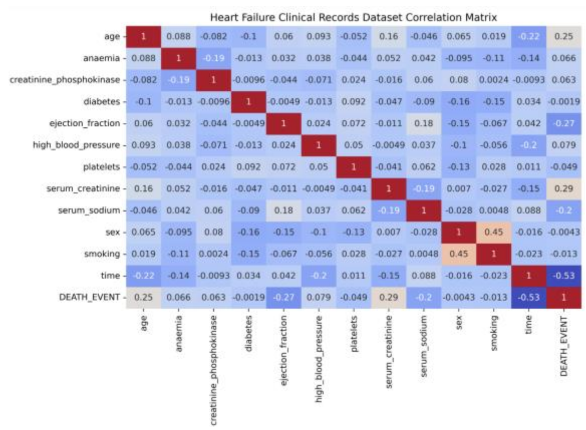
FIGURE 2.1.2: SUBSET OF DISTRIBUTION PLOTS

A minimum body fat percentage of 0% was indicated in the summary statistics and was removed as it is not biologically plausible (Matthews, 2020).

Figure 2.1.1 presents the correlation matrix, highlighting significant correlations among variables. Body fat has a strong positive correlation with many body circumference measurements, indicating that as these measurements increase, body fat percentage also increases. On the other hand, Height and Density has a negative correlation with body fat, indicating that as these variables decrease, body fat percentage increases. Since Density has an almost perfect correlation with body fat and that it is used to estimate body fat percentage (Body Fat Prediction Dataset, 2021), it will be removed to prevent multicollinearity.

Figure 2.1.2 presents a subset of the 28 histograms and boxplots due to constraints. Right-skewed distributions are predominant across the variables, and clear outliers can be observed in the boxplots. Therefore, outliers will be removed to minimise the influence of extreme values and improve the model's performance.

2.2 Heart Failure Clinical Records Dataset



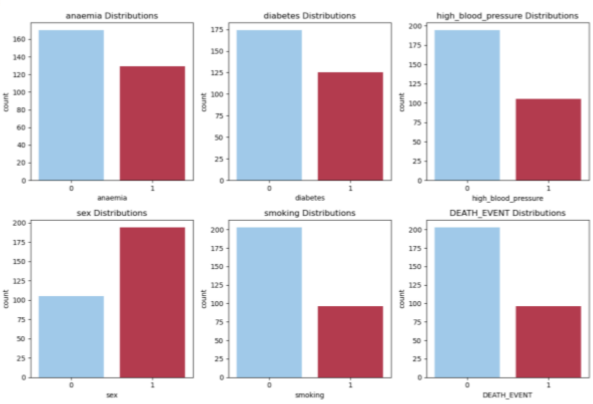


FIGURE 2.2.1: CORRELATION MATRIX

FIGURE 2.2.2: COUNT PLOT

Figure 2.2.1 presents the correlation matrix of heart failure data, revealing weak linear relationships between Death Event and other variables. This suggests that individual variables may not strongly influence the outcome of Death Event. However, Time shows a moderate negative correlation, indicating that as time progresses, individuals tend to have a lower risk of death.

Figure 2.2.2 presents a count plot of the categorical binary variables. It shows that the target variable is imbalanced, which may result in class imbalance in the training and test sets as well. This leads us to use *F1-score* and *balanced accuracy score* as our evaluation metrics.

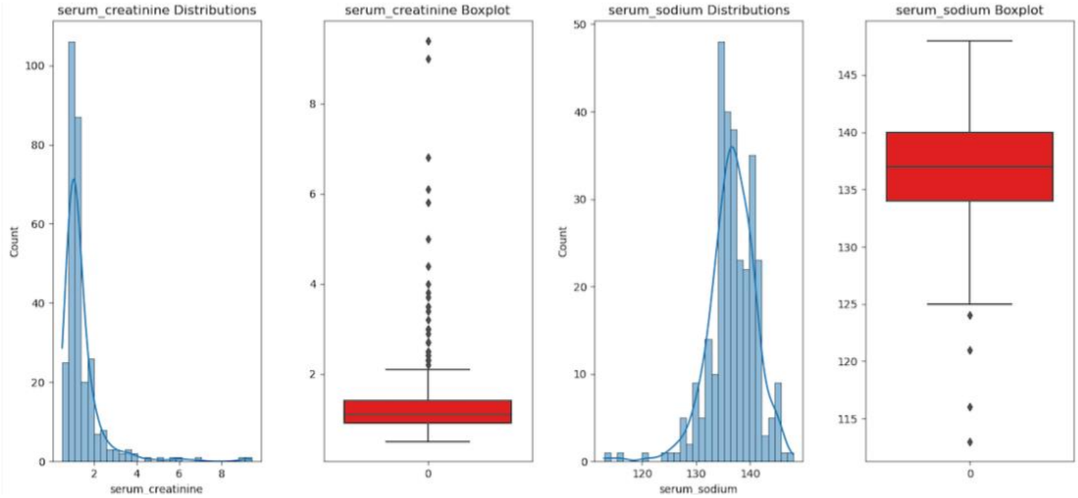


FIGURE 2.2.3: SUBSET OF DISTRIBUTION PLOTS

Figure 2.2.3 presents a subset of the 14 histograms and boxplots of continuous variables. Right-skewed distributions are also predominant across the variables, and outliers are evident in the boxplots. As a result, it will be removed to minimise the impact of extreme values and increase model accuracy.

3. Unsupervised Learning

3.1 Research Questions

Research questions (RQ) addressing substantive issues in the body fat data:

- RQ1: What is the optimal number of clusters for assessing body fat percentage?
- RQ2: What do the optimal clusters represent?

3.2 Principal Component Analysis

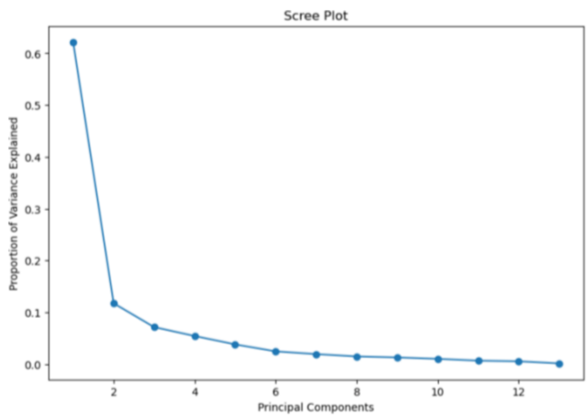


FIGURE 3.2: SCREE PLOT

Principal Component Analysis (PCA) is a dimensionality reduction technique that is used to reduce a dataset with highly correlated variables into a dataset of uncorrelated variables while maximising the variance of the dataset. The reduced uncorrelated variables are called principal components (PC). The maximisation of variance allows the preservation of patterns and relationships in the data (GeeksforGeeks, 2025). Data preprocessing is done before applying PCA. The target variable, 'Body Fat' is removed and all remaining variables are standardized to ensure a mean of zero and a standard deviation of one. Figure 3.2 presents a scree plot which indicates the proportion of overall variance accounted for by each principal component. To determine the optimal number of components needed, the elbow method is used where a 'bent' in the plot indicates the point where the variance explained reduces sharply. Therefore, two principal components were chosen. These principal components will then be used for clustering, improving cluster separation by reducing redundancy.

3.3 K-means Clustering

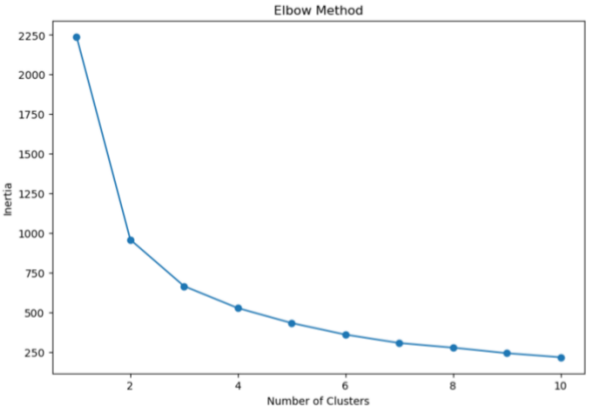


FIGURE 3.3: ELBOW METHOD

K-means Clustering is an unsupervised clustering technique which groups the data into similar clusters. The number of clusters, K, is specified first before categorising the data points into those clusters. K number of centroids are initialised, before allocating each point to the nearest centroid. The centroid is then recalculated, and for a certain number of iterations, the process is repeated until the centroid doesn't change significantly (GeeksforGeeks, 2025). The elbow method was used to identify the optimal number of clusters. Based on the 'bent' in Figure 3.3, two clusters were identified. The two clusters were then visualised in Figure 3.5.1.

3.4 Hierarchical Clustering

Hierarchical Clustering is also an unsupervised clustering technique which groups data into similar clusters. Unlike K-means, there is no need to specify the number of clusters first. Instead, in the Agglomerative Clustering approach, each data point will be considered as its individual cluster. Then, distances between clusters are calculated through Euclidean Distancing, and the two closest clusters are merged. This continues until only one cluster is left (GeeksforGeeks, 2025). To determine the optimal number of clusters, the longest vertical lines between clusters are located, and a horizontal line is plotted. The number of vertical lines crossed by this horizontal line represents the number of optimal clusters (Sampaio, 2023). Based on Figure 3.4, two clusters were identified. The two clusters were then visualised in Figure 3.5.2.

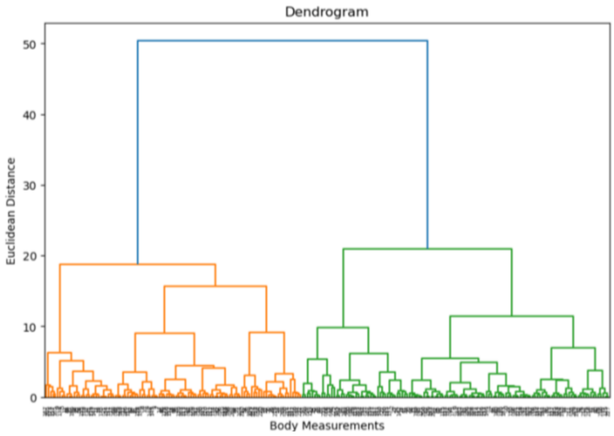


FIGURE 3.4: DENDROGRAM

3.5 Results

Figure 3.5.1 and 3.5.2 shows two distinct well-separated clusters along PC1, implying the existence of two body composition groups. Thus, the optimal number of clusters for assessing body fat percentage is two.

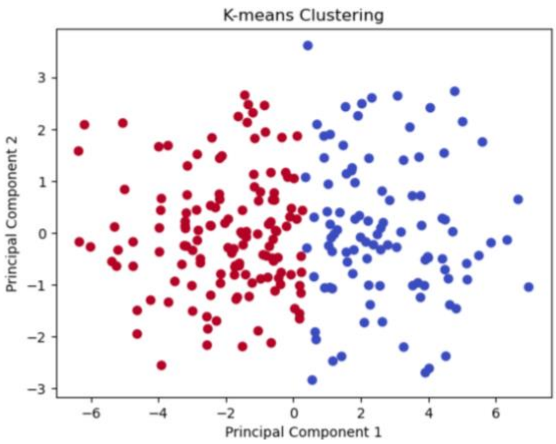


FIGURE 3.5.1: CLUSTER PLOT OF K-MEANS

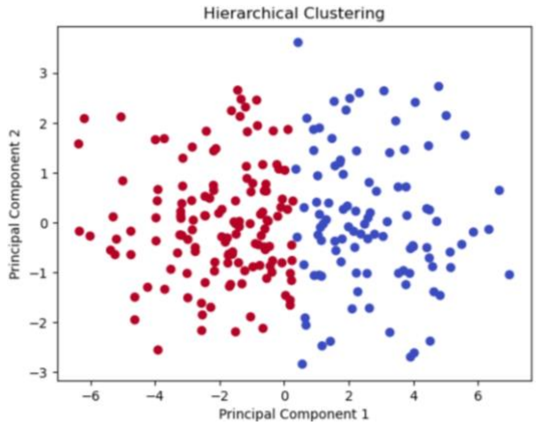


FIGURE 3.5.2: CLUSTER PLOT OF HIERARCHICAL

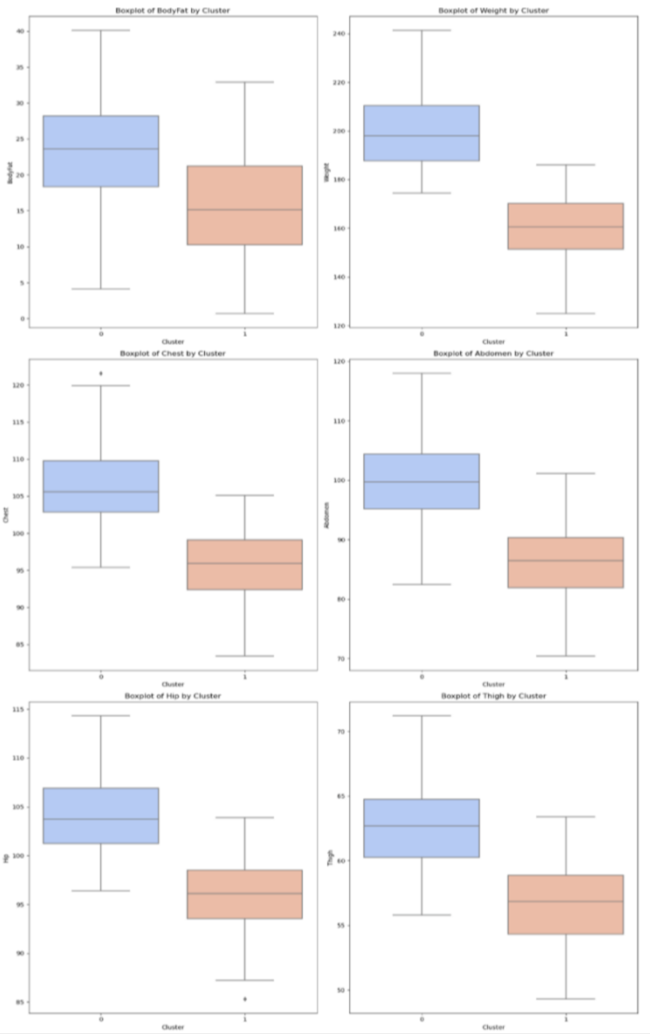


FIGURE 3.5.3: CLUSTER ANALYSIS

To understand the key difference between the two clusters, the average values of all features were compared.

Features that showed a difference of more than four units between the clusters were selected for further analysis. These features are – Body Fat Percentage, Weight, Chest, Abdomen, Hip and Thigh, where they exhibited the most significant separation between Cluster 0 and Cluster 1. Figure 3.5.3 shows boxplots for these features, and it can be seen that cluster 0 represents the cluster with higher body fat percentage, while cluster 1 represents the cluster with lower body fat percentage. Individuals in cluster 0 tend to have higher weight, larger chest, abdomen, hip and thigh measurements on average as compared to individuals in cluster 1. These findings suggests that clustering can effectively differentiate individuals based on body fat and their physique, making it useful for medical assessments and the tailoring of fitness programs according to body types.

4. REGRESSION

Due to the multicollinearity detected in the correlation matrix (Figure 2.1.1), Lasso Regression was chosen for feature selection and to handle multicollinearity. It is a regularisation technique where a penalty term (*lambda*) is added which minimises the sum of squared differences. It shrinks coefficients of insignificant variables to zero, reducing overfitting. The optimal lambda is chosen using cross-validation (GeeksforGeeks, 2025). A 70-30 train-test split was done on the original body fat data, followed by scaling and cross-validation. An optimal lambda of 0.1 was generated, and feature selection was performed. The selected optimal features are as follows:

```
Optimal Features:
['Age', 'Height', 'Neck', 'Chest', 'Abdomen', 'Thigh', 'Knee', 'Ankle', 'Wrist']
```

FIGURE 4: OPTIMAL FEATURES FOR REGRESSION

Moving forward, the dataset will be reduced to the selected features, which will be used in the regression models. To prepare for Linear, Lasso and Ridge Regression, the new modified dataset was scaled to ensure consistency. No scaling was done for Decision Tree and Random Forest as it is not required for tree models.

4.1 Research Questions

Research questions (RQ) addressing substantive issues in the body fat data:

- RQ1:Which body measurements are the most significant predictors of body fat percentage?
- RQ2: How do different regression models compare in predicting body fat percentage?

4.2 Linear Regression

```
Linear Regression MSE: 23.261
Linear Regression RMSE: 4.823
Linear Regression MAE: 4.015
Linear Regression R-Squared: 0.603
```

FIGURE 4.2: EVALUATION RESULTS OF LINEAR REGRESSION

Linear Regression is a statistical model utilised to analyse the relationship between a continuous dependent variable and one or more independent variables. The model estimates this relationship by fitting a linear equation with actual data. It then uses this equation to predict the dependent variable using independent variables (GeeksforGeeks, 2025). Linear regression was applied to the modified data, and its performance was evaluated using an array of evaluation metrics to assess model accuracy. The evaluation results are shown.

4.3 Lasso Regression

```
Lasso Regression MSE: 22.760
Lasso Regression RMSE: 4.771
Lasso Regression MAE: 4.012
Lasso Regression R-Squared: 0.611
```

FIGURE 4.3: EVALUATION RESULTS OF LASSO REGRESSION

Lasso Regression was applied to the modified data, and the tuned optimal alpha of 0.1 was used. The evaluation results of lasso regression are as shown.

4.4 Ridge Regression

```
Ridge Regression MSE: 23.197
Ridge Regression RMSE: 4.816
Ridge Regression MAE: 4.018
Ridge Regression R-Squared: 0.604
```

FIGURE 4.4: EVALUATION RESULTS OF RIDGE REGRESSION

Ridge Regression is also a regularisation technique where a penalty term (*lambda*) is added to minimise the sum of squared differences. However, instead of shrinking coefficients of insignificant variables to zero, it makes the coefficient smaller and closer to zero (GeeksforGeeks, 2025). The optimal lambda, also selected via cross-validation, was found to be 1.0. Ridge regression was applied to the modified data using the optimal lambda, and the evaluation results are as shown.

4.5 Decision Tree Regression

```
Pruned Decision Tree MSE: 29.928
Pruned Decision Tree RMSE: 5.471
Pruned Decision Tree MAE: 4.405
Pruned Decision Tree R-Squared: 0.489
```

FIGURE 4.5: EVALUATION RESULTS OF DECISION TREE REGRESSION

Decision Tree Regression is a non-linear regression model that uses a tree-base method to derive predictions. It divides the data into smaller subsets based on input features, and it is to minimise the predicting error of the target variable (GeeksforGeeks, 2025). Pre-pruning is done via cross-validation to find the best parameters before fitting the modified model. To identify if post-pruning is needed, the Mean Squared Error (MSE) for the training and test data was computed. Since the training MSE is lower than the test MSE, it suggested overfitting and post-pruning was done via cost complexity pruning, which was also determined using cross-validation. Decision tree regressor was applied to the modified data again, and the evaluation results are as shown.

4.6 Random Forest Regression

```
Random Forest MSE: 22.140
Random Forest RMSE: 4.705
Random Forest MAE: 3.914
Random Forest R-Squared: 0.622
```

FIGURE 4.6: EVALUATION RESULTS OF RANDOM FOREST REGRESSION

Random Forest Regression is an ensemble learning method that merges several decision trees to generate a more reliable prediction. It creates multiple decision trees during training, where each tree is trained on a random sample of the data (GeeksforGeeks, 2025). Cross-validation was done to find optimal parameters, where it was used on the modified data. The evaluation results are as shown.

4.7 Model Evaluation

The evaluation metrics used are Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and R-Squared. MSE, RMSE and MAE calculates the difference between the predicted and actual values, each in their unique ways, to assess the accuracy of the model's predictions. Lower values indicate better model performance. R-squared calculates how much variance the model can explain, in the range of zero to one. Higher values closer to one indicates better fit and explains more of the variance in the data (GeeksforGeeks, 2025).

From Figures 4.2 to 4.6, it can be observed that Ridge and Linear Regression showed similar results with an RMSE of 4.82, MAE of 4.01 and R-Squared of 0.60, while the pruned Decision Tree had the worst performance with an RMSE of 5.47, MAE of 4.40 and R-Squared of 0.49. On the other hand, Lasso Regression exhibited the best performance, demonstrating a lower RMSE of 4.77 and a higher R-Squared of 0.61. This suggests that Lasso Regression is the most accurate model among the ones used, and has the better potential to predict body fat percentage. Additionally, the results of feature selection suggests that Age, Height, Neck, Chest, Abdomen, Thigh, Knee, Ankle and Wrist are the most significant predictors of body fat percentage as they have non-zero coefficients in the Lasso model. Among these, Abdomen has the largest positive coefficient which can highlight its importance in accurately predicting body fat percentage. It could be used in future studies by researchers as a key variable, or in health assessments as a non-invasive procedure.

5. Classification

Random Forest Classifier's built-in feature importance was used to perform feature selection to identify the most optimal variables. Similar to the Random Forest Regressor, the classifier constructs many decision trees but it is used for predicting categorical outcomes instead (GeeksforGeeks, 2025). A 70-30 train-test split was done on the heart failure data, followed by hyperparameter tuning via cross-validation to obtain the best parameters. The classifier was then fitted onto the ideal parameters, generating the selected features via feature importance, as shown:

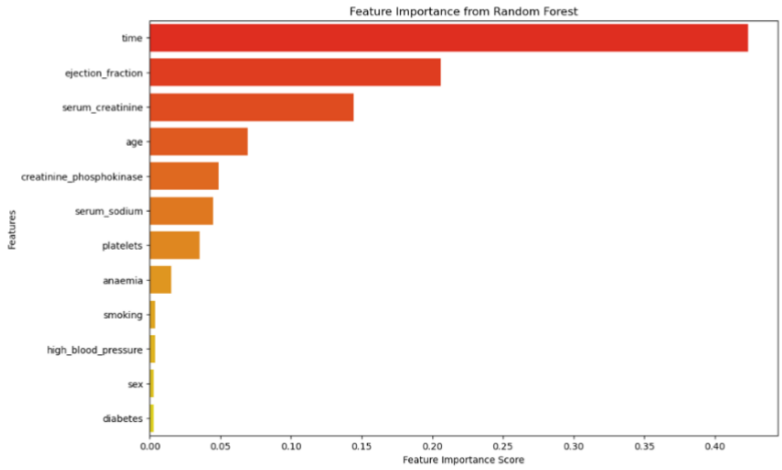


FIGURE 5: FEATURE IMPORTANCE OF VARIABLES

Moving on, the top four optimal features were selected, and the heart failure data will be reduced to these features. The modified data will be used in the following classification models. To prepare for Logistic Regression and K-Nearest Neighbours, the new modified dataset was scaled to ensure consistency. No scaling was done for Decision Tree and Random Forest as it is not required for tree models.

5.1 Research Questions

Research questions (RQ) addressing substantive issues in the heart failure data:

- RQ1: Which clinical or lifestyle features are the strongest predictors of mortality in heart failure patients?
- RQ2: How do different classification models compare in predicting mortality?

5.2 Logistic Regression

Logistic Regression, is a variation of Linear Regression but it is used for predicting categorical outcomes instead. It uses a sigmoid function to produce a probability value of zero to one, and assigns data points to different classes based on a threshold value (GeeksforGeeks, 2025). Cross-validation was performed to select the best parameters while ensuring compatibility between penalties and solvers. Logistic regression was then applied using the optimal parameters, and evaluation metrics were generated for each configuration. The best evaluation results are as shown.

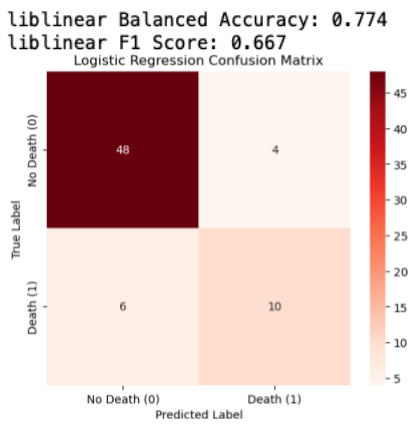


FIGURE 5.2: EVALUATION RESULTS OF LOGISTIC REGRESSION

5.3 K-Nearest Neighbours

K-Nearest Neighbours (KNN) classifies a point by considering the most frequent class among its nearest points (neighbours). The value ‘K’ determines how many nearby points are considered when making a prediction. It then classifies the point according to the majority class of its 'K' nearest neighbours (GeeksforGeeks, 2025). The optimal parameters was chosen via cross-validation, before applying KNN onto the modified data. The evaluation results are as shown.

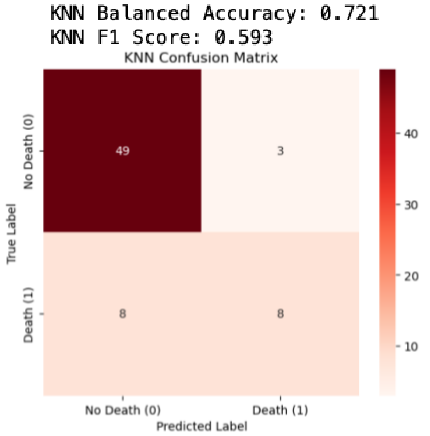


FIGURE 5.3: EVALUATION RESULTS OF KNN

5.4 Decision Tree Classification

Decision Tree Classification is similar to the regression where both models follow the same tree-based structure, except that it classifies categorical outcome instead. Pre-pruning was also done via cross-validation to find the best parameters before fitting the modified model. To identify if post-pruning is needed, the balanced accuracy for the training and test data was computed. Since the training balanced accuracy is higher than the test balanced accuracy, it suggested overfitting and post-pruning was done via cost complexity pruning, which is also determined using cross-validation. Decision tree classifier was applied to the modified data again, and the evaluation results are as shown.

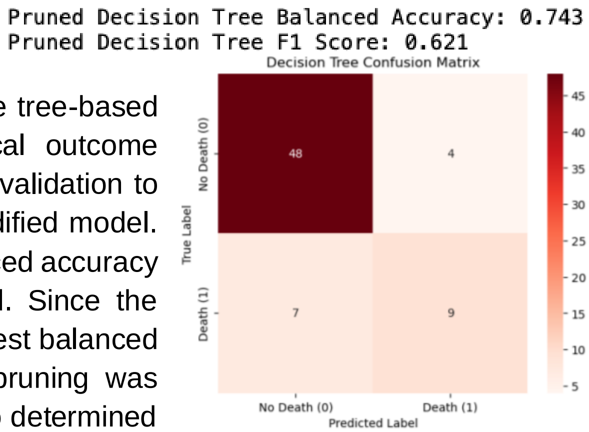


FIGURE 5.4: EVALUATION RESULTS OF DECISION TREE CLASSIFICATION

5.5 Random Forest Classification

Random Forest Classifier was applied to the modified data, using the optimal parameters. The evaluation results are as follows:

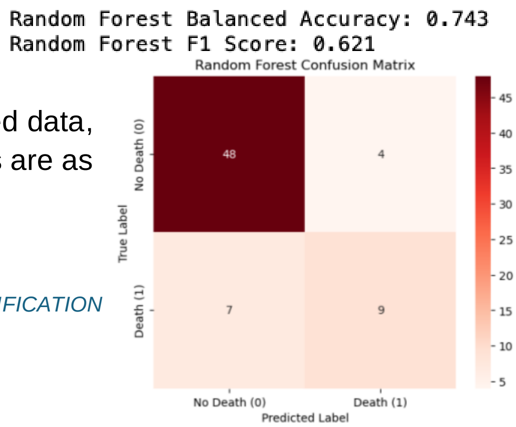


FIGURE 5.5: EVALUATION RESULTS OF RANDOM FOREST CLASSIFICATION

5.6 Model Evaluation

The evaluation metrics used are Balanced Accuracy, F1-Score and Confusion Matrix. Balanced accuracy adjusts for class imbalance by averaging Sensitivity and Specificity. It provides a better measure of performance when the dataset has an unequal class distribution. F1-Score is the weighted mean of Precision and Recall, balancing false positives and false negatives. For both metrics, higher values indicate better model performance. Confusion matrix visualises how many values were correctly or incorrectly classified, and it is used to calculate balanced accuracy and f1-score (Kiecodes, 2024).

From Figures 5.2 to 5.5, it can be observed that pruned Decision Tree and Random Forest had the same results for both balanced accuracy and f1-score – 0.743 and 0.621 respectively. KNN had the worst performance with a balanced accuracy of 0.721 and f1-score of 0.593. On the other hand, Logistic Regression exhibited the best performance of a higher balanced accuracy and f1-score – 0.774 and 0.667 respectively. Additionally, while all confusion matrixes have the same output for True Positives, Logistic Regression has the highest output for True Negatives. This suggests that Logistic Regression is the most accurate model among the ones used, and has the better potential to classify mortality. Additionally, the selected features – Time, Ejection Fraction, Serum Creatinine and Age are the most significant predictors of mortality as they have the highest importance scores. Among these, Time holds the highest importance score which can highlight its importance in accurately classifying mortality. Future studies could focus on Time as a key variable in understanding mortality risk, potentially exploring its relationship with treatment outcomes and condition progression.

6. RELATED LITERATURE AND CONCLUSION

Literature by Fan, Chiong, Hu, Keivanian and Chiong (Fan et al., 2022) discussed <u>body fat prediction</u> and found that the models exhibited better performance when feature extraction was used. Their research used *Multilayer Perceptron*, *Support Vector Machine*, *Random Forest* and *XGBoost* alongside feature extraction methods *Factor Analysis*, *Principal Component Analysis* and *Independent Component Analysis* which differs from this research's models and feature extraction method. In this research, an average RMSE and MAE of 4.92 and 4.07 was observed, which aligned with their results. One possible reason for this could be due to feature extraction, which suggests that feature extraction methods can enhance model performance by reducing dimensionality and capturing important patterns in the data, and highlights the importance of feature extraction which could be explored further for improvement in future studies.

Additionally, literature by Chicco and Jurman (Chicco & Jurman, 2020) discussed <u>predicting survival of patients with heart failure</u> and found that serum creatinine and ejection fraction are the two most relevant features. Their research used ten different models which differs from this research. In this research, additional Time and Age was observed as relevant features, which is in contrast with their findings as Time held the highest importance score. One possible reason for this difference could be variations in feature selection methods, which may have emphasised different aspects of the data. This suggests that feature importance can vary based on the analysis models, emphasising the need to explore how different factors contribute to mortality.

In conclusion of this research, unsupervised learning, regression and classification techniques were constructed in analysing body fat prediction and heart failure mortality. PCA and clustering techniques revealed distinct patterns in the body fat data, and regression models identified significant predictors. Meanwhile, classification models identified significant predictors in the heart failure data. Future research could explore additional techniques to promote health management.

7. References

Body fat Prediction Dataset. (2021, June 14). Kaggle. https://www.kaggle.com/datasets/fedesoriano/body-fat-prediction-dataset/data

British Heart Foundation. (n.d.). Best way to measure body fat. https://www.bhf.org.uk/informationsupport/heart-matters-magazine/nutrition/weight/best-way-to-measure-body-fat

Chicco, D., & Jurman, G. (2020). Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone. *BMC Medical Informatics and Decision Making*, *20*(1). https://doi.org/10.1186/s12911-020-1023-5

Fan, Z., Chiong, R., Hu, Z., Keivanian, F., & Chiong, F. (2022). Body fat prediction through feature extraction based on anthropometric and laboratory measurements. *PLoS ONE*, *17*(2), e0263333. https://doi.org/10.1371/journal.pone.0263333

GeeksforGeeks. (2025, February 3). *Principal Component Analysis(PCA)*. https://www.geeksforgeeks.org/principal-component-analysis-pca/

GeeksforGeeks. (2025, January 15). *K means Clustering Introduction*. https://www.geeksforgeeks.org/k-means-clustering-introduction/

GeeksforGeeks. (2025, February 4). *Hierarchical clustering in machine learning*. https://www.geeksforgeeks.org/hierarchical-clustering/

GeeksforGeeks. (2025, March 27). *What is Lasso Regression?* https://www.geeksforgeeks.org/what-is-lasso-regression/

GeeksforGeeks. (2025, January 16). *Linear Regression in Machine learning*. https://www.geeksforgeeks.org/ml-linear-regression/

GeeksforGeeks. (2025, February 12). *Ridge regression*. https://www.geeksforgeeks.org/what-is-ridge-regression/

GeeksforGeeks. (2025, January 29). *Python | Decision Tree Regression using sklearn*. https://www.geeksforgeeks.org/python-decision-tree-regression-using-sklearn/

GeeksforGeeks. (2025, February 3). *Random forest regression in Python*. https://www.geeksforgeeks.org/random-forest-regression-in-python/

GeeksforGeeks. (2025, March 11). *Random Forest Classifier using Scikitlearn*. https://www.geeksforgeeks.org/random-forest-classifier-using-scikit-learn/

GeeksforGeeks. (2025, February 3). *Logistic regression in machine learning*. https://www.geeksforgeeks.org/understanding-logistic-regression/

GeeksforGeeks. (2025, January 29). *KNeArest Neighbor(KNN) algorithm*. https://www.geeksforgeeks.org/k-nearest-neighbours/

Heart Failure Clinical Records. (2020). [Dataset]. https://doi.org/10.24432/C5Z89R

Kiecodes. (2024, February 22). Five Machine Learning Metrics every Engineer should know. *Medium*. https://kiecodes.medium.com/ml-metrics-33687c05425f

Matthews, M. (2020, July 8). Why it's not humanly possible to have less than 1 percent body fat. *Men's Health*. https://www.menshealth.com/health/a33247811/ronnie-coleman-body-fat/

Otis, B. O. (2025, January 17). *Body fat - the nutrition source*. The Nutrition Source. https://nutritionsource.hsph.harvard.edu/healthy-weight/measuring-fat/

Sampaio, C. (2023, November 16). *Definitive Guide to Hierarchical Clustering with Python and Scikit-Learn*. Stack Abuse. https://stackabuse.com/hierarchical-clustering-with-python-and-scikit-learn/

Website, N. (2024, April 8). *Heart failure*. nhs.uk. https://www.nhs.uk/conditions/heart-failure/

What is Heart Failure? (2023, March 22). American Heart Association. https://www.heart.org/en/health-topics/heart-failure/what-is-heart-failure

What is Machine Learning? Definition, Types, Tools & More. (2024, November 8). https://www.datacamp.com/blog/what-is-machine-learning

8. Appendix

8.1 Body Fat Prediction Dataset

Variables	Description
Density	Gram per cubic centimetre (gm/cm^3)
Body Fat Percentage	Percentage (%)
Age	Years
Weight	Pounds (Lbs)
Height	Inches
Neck Circumference	Centimetre (cm)
Chest Circumference	Centimetre (cm)
Abdomen 2 Circumference	Centimetre (cm)
Hip Circumference	Centimetre (cm)
Thigh Circumference	Centimetre (cm)
Knee Circumference	Centimetre (cm)
Ankle Circumference	Centimetre (cm)
Biceps (Extended) Circumference	Centimetre (cm)
Forearm Circumference	Centimetre (cm)
Wrist Circumference	Centimetre (cm)

8.2 Heart Failure Clinical Records Dataset

Variables	Description
Age	Years
Anaemia	0 = No anaemia 1 = Anaemia
Creatinine Phosphokinase	Level of CPK enzyme in the blood
Diabetes	0 = No diabetes 1 = Diabetes
Ejection Fraction	Percentage of blood leaving
High Blood Pressure	0 = No high blood pressure 1 = High blood pressure
Platelets	Level of platelets in the blood
Serum Creatinine	Level of creatinine in the blood
Serum Sodium	Level of sodium in the blood
Sex	0 = Woman 1 = Man
Smoking	0 = Non-smoker 1 = Smoker
Time	Follow-up period
Death Event	0 = Patient did not die during the follow-up period 1 = Patient died during the follow-up period